

Edible Light Pipe Made of Candy

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ABSTRACT

In recent years, there have been widespread attempts to enhance and extend the sense of food using light and images. As one of such attempts, a method to create transparent optical elements using foods such as candy and agar has been proposed. In this study, an edible light pipe made of candy is proposed as a new edible optical element to decorate foods by light. We report the results of basic evaluation of a prototype light pipe made of candy. We also propose a method of supplying light to the light pipe from a remote location using a laser.

Keywords: Light Pipe, Edible, Food, Laser.

Index Terms: Hardware—Emerging technologies—Analysis and design of emerging devices and systems—Emerging tools and methodologies

1 INTRODUCTION

There have been many attempts to enhance and extend the dining experience by adding images to food. Examples of attempts to project images onto tables and food include the Dinning Presenter [1], which projects images onto tables, and the Spot-Light [2], which projects images and a sound field onto food.

In addition, a method to add optical functions by molding transparent food materials has recently been proposed [3, 4]. For example, edible retroreflectors made from agar and candy have been reported [5, 6]. Among the many other functions that can be realized by molding transparent foodstuffs, we focused on light pipes.

As shown in the figure 1, a light pipe is an optical element that uses the property of total reflection to transmit light incident at the entrance to the exit. It can also be used as a kind of illumination by scattering light propagated inside by contacting the surface with an object or intentionally creating uneven surfaces. By combining food and light pipes, it is expected that new methods of food presentation using light will become possible. In addition, if light pipes are made from foodstuffs, they can be used safely and harmlessly on food.

In this paper, we propose a light pipe made of candy, which is a transparent food material, and report on the method of making the light pipe and the results of verifying its potential applications.



Figure 1: Light pipe

2 PROPOSAL FOR EDIBLE LIGHT PIPE

2.1 Prototype candy light pipe

In this study, edible light pipes are produced using reduced isomaltulose, which is characterized by its transparency and resistance to

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caramelization even when boiled down to high temperatures. Candy light pipes were made by one of the following two methods.

2.2 Candy light pipe made with molds

For light pipes with complex shapes, candy light pipes were fabricated using molds. In this paper, a light pipe with the shape of a candle and a light pipe with the function of bending light by 90 degrees were selected as prototype objects. The following is the procedure for making a candy light pipe using the mold. A schematic diagram of the fabrication procedure is shown in Figure 2.

1. Fix a light pipe that serves as a mold to the container and pour in food-grade silicone. When the silicone hardens, remove it from the container and make an incision to a depth of 2/3.
2. Put reduced isomaltulose and water into the silicone mold and heat it in a microwave oven until the candy melts.
3. Since the candy boils and bubbles are formed during heating, the silicone mold is placed in a vacuum chamber with a hot plate set at 170°C and the bubbles are removed while reducing the pressure.
4. Cool the candy at room temperature, and remove the candy from the silicone mold when it is hard enough.

The commercial light pipe used as a mold is shown on the left in Figure 3 and the candy light pipe made according to the above procedure is shown on the right in Figure 3.

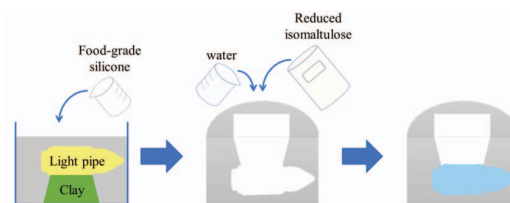


Figure 2: Fabrication procedure using moulds



Figure 3: commercially available light pipe & Candy light pipes using models

2.3 Tubular candy light pipes

Tube-shaped light pipes are fundamental for transmitting light and can be easily formed using commercially available tubes as molds. Therefore, we fabricate candy light pipes using commercially available tubes. The fabrication procedure for making tubular candy light pipes is described below. A schematic diagram of the fabrication procedure is shown in Figure 4. A photograph of a candy light pipe

actually fabricated by the procedure described above is shown in Figure 5.

1. Attach a cork wrapped in plastic wrap to one end of the tube and a funnel to the other.
2. Put the reduced isomaltulose and water in the pan, mix with a rubber spatula, and heat the candy to 170°C.
3. Cool the candy until it stops boiling, pour the candy into the tube that has been warmed enough, and plug it with a cork wrapped in plastic wrap.
4. Fix the tube containing the candy into the desired shape and cool the candy at room temperature. When it hardens until it is no longer deformed, cut a slit in the tube with a cutter and take out the candy.

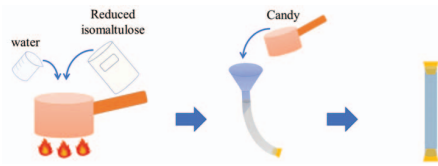


Figure 4: Fabrication procedure using tubes



Figure 5: Silicon tube & Tubular candy light pipes

2.4 Evaluation of tubular candy light pipes

We evaluated the ability of a tubular candy light pipe to transmit light to a remote location using the following method. The light intensity of light coming from the light source with and without the tubular candy light pipe in front of the light source was captured by a camera, and by comparing the pixel values of the images, it was confirmed that the candy light pipe can transmit light to remote locations. The environment during photography is shown in Figure 6. A USB camera, a piece of cardboard with a 6 mm diameter hole, and a light source (light guide) were placed in a straight line at 15 cm intervals. Images of light arriving from the light source were captured under two conditions: with and without a tubular candy light pipe placed between the perforated cardboard and the light source. The image taken with the candy light pipe placed is shown on the left side of Figure 7, and the image taken without the pipe is shown on the right.

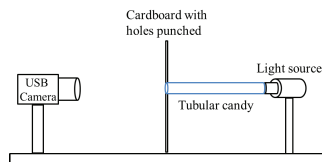


Figure 6: Experimental environment

We averaged the pixel values from this image only over the area where the light was present. The average pixel value was 131.52 when the candy light pipe was placed, and 97.79 when it was not placed. As a result, it was confirmed that the candy light pipe improved the efficiency of light transmission.



Figure 7: Experimental results

3 APPLICATION EXPERIMENT

In this paper, we propose a staging technique to make jellies glow as a way of utilizing candy light pipes. When the jelly and the light pipe come into contact with each other, the condition of total reflection of the light pipe is broken at the point of contact, and light is refracted and passes through the jelly. By utilizing this property, it is possible to make the jelly glow using the light pipe. The figure 8 on the right shows how the jelly is illuminated by using a light guide to insert light into a tubular light pipe.

If the light source is placed near the light pipe, it is necessary to place inedible items such as LEDs near the light pipe, but this would lose the advantage of being able to safely use the light pipe on top of the edible light pipe food. To solve this problem, we propose a method that uses lasers to introduce light from a remote location. In this method, edible retroreflectors placed near the light pipe are used as a reference, and a galvano mirror is used to control the direction of the laser beam. The left side of the figure 8 shows how the jelly is illuminated by this method.

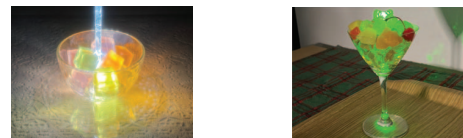


Figure 8: Photographs of glowing jelly.

4 CONCLUSION

In this study, we fabricated an edible light pipe using candy, a transparent food material. By evaluating the light pipe made of candy by pixel value, we confirmed that it can transmit light as efficiently as a conventional light pipe. We also proposed a glowing dish using the light pipe made of candy and a method of providing light from a distance.

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REFERENCES

- [1] M. Mori, K. Kurihara, K. Tsukada, and I. Sioo. Dining presenter: Augmented reality system for a dining tabletop. In *Proc. 11th Ubicomp*, pp. 168–169. Orlando, 2009.
- [2] Y. Kita and J. Rekimoto. Spot-light: Multimodal projection mapping on food. In *Proc. HCI International 2013 - Posters' Extended Abstracts*, pp. 652–655, 2013. doi: 10.1007/978-3-642-39476-8_131
- [3] E. Fujiwara, M. Cabral, Thiago D. and Sato, H. Oku, and C. M. B. Cordeiro. Agarose-based structured optical fibre. *Scientific Reports*, 10(1):7035, Apr 2020. doi: 10.1038/s41598-020-64103-3
- [4] M. Nomura and H. Oku. Edible lens made of agar. *Optical Review*, 27(1):9–13, Feb 2020. doi: 10.1007/s10043-019-00557-9
- [5] H. Oku, T. Uji, Y. Zhang, and K. Shibahara. Edible fiducial marker made of edible retroreflector. *Computers & Graphics*, 77:156–165, 2018. doi: 10.1016/j.cag.2018.10.002
- [6] H. Oku, M. Sato, and Y. Funato. Edible retroreflector made of candy. *IEEE Access*, 10:24749–24758, 2022. doi: 10.1109/ACCESS.2022.3155603